AP PHYSICS 1 IA-1 — COMPLETE EXTRACTED DATA (Public Archive)

Everything extracted from the three sources: **Newton School Portal**, **Slack**, **WhatsApp hardcopy photos (146)**.

Compiled 26 Oct 2026. Exam: Applied Physics 1 IA-1, Tue 27 Oct 2026, 9:40 AM.

SOURCE 1 — PORTAL (my.newtonschool.co, course tjj12ybxo9tt)

Course: NSTB'26-CS, subject **AP-Robo B** = Applied Physics 1. Parent course kxz81gy1l9wm.

My performance (Overview): Lectures 83% (10/12) · Assignments 0% (0/0) · Contests 50% (1/2) · Quizzes 0% (0/0). Total XP 5,795.

Lectures (13 total):

#	Date	Topic	Status
L1	31 Aug	Measurement Fundamentals	attended
L2	2 Sep	Scalars and Vectors, Vector Arithmetic, Vector Products, Vector Resolution	attended
L3	7 Sep	Kinematics, Distance and Displacement, Speed and Velocity, Acceleration	attended
L4	9 Sep	Relative Motion, Solving 1-D Motion, Projectile Motion, Breaking Down Multi-D Motion	attended, notes
L5	16 Sep	NLMs & FBDs, Forces & Types, Newton's Laws, Frames of Reference, Pseudo Forces	notes, VIDEO UNWATCHED (0/30)
L6	21 Sep	Friction, Normal & Coefficient of Friction, Friction & Types, Multi-D FBDs	attended, notes
L7	23 Sep	Circular Motion, Angular Coordinate System, Radial/Tangential Vectors	VIDEO UNWATCHED (0/30)
L8	28 Sep	Work and Energy	attended
L9	30 Sep	Circular Motion / CM	attended, notes
L10	5 Oct	Circular Motion / CM	attended, notes
L11	7 Oct	(Circular Motion)	VIDEO UNWATCHED (0/30)
L12	12 Oct	(Circular Motion)	attended
L13	14 Oct	(Circular Motion)	attended

Note: **IA-1 syllabus stops at L10.** L11–13 are outside the exam.

Contests:

- Assignment Contest 1 — solved (1/2 counted).
- **Applied Physics 1 - Contest 2 (Subjective)** — released 16 Oct, deadline 16 Oct 11:20 AM, topic **Centre of Mass**, difficulty easy, **0/40 XP earned**, solved by 34 students (15%). STILL OPEN BACKLOG.

Assignments: none found (0/0). **Quiz:** 0/0 (1 historical quiz contest, completed).

Contest-2 question text (playground/subjective/9gkqeb7ttnk6): an ice cube (mass m, side L) melts in a tray (mass M); the water spreads into a thin layer. Does the centre of mass of the system move? Explain without

numerical calculation: how the redistribution of mass affects it, and how the relative masses m and M influence the magnitude of the change.

SOURCE 2 — SLACK (#academic-announcement-26)

Official AP 1 IA-1 syllabus (nikhil.menon, 14 Oct 2026, 5:14 PM) — VERBATIM, "From Lecture 1 – Lecture 10":

Sizes in the Universe · Introduction to Measurement · Units and Dimensions · Order of Magnitude · Significant Digits · Scientific Notation · Error and Propagation · Scalars and Vectors · Vector Arithmetic · Vector Products · Vector Resolution · Differentiation · Integration · Distance and Displacement · Speed and Velocity · Acceleration and Jerk · Graphical Understanding of Motion · Motion in Straight Line · Equations of Motion · Relative Motion · Solving 1-D Motion · Projectile Motion · Breaking Down Multi-D Motion · Forces & Types · Newton's Laws of Motion · Frames of Reference · Pseudo Forces & Inertia · Free Body Diagrams · Friction & Types · Normal & Coefficient of Friction · Multi-D FBDs with Friction · Multi-D NLM & Friction Problems · Angular Coordinate System · Radial and Tangential Vectors · CM: Angular Displacement & Speed · CM: Angular Velocity & Acceleration · CM: Centripetal & Centrifugal · Solving Circular Motion · Kinetic Energy · Potential Energy · Work Done · Spring Force · Conservative vs Non-Conservative Forces · Work and Energy Theorem · Centre of Mass · CoM: System of Objects · CoM: Distributed Systems · CoM: Multi-Dimensional Objects · Linear Momentum · Conservation of Linear Momentum · Conservation of LM: Multi-Dimensional · Conservation of LM: Multi-Stage · Collisions for Rigid Bodies · Elastic Collision · Inelastic Collision · Coefficient of Restitution · Multi-Dimensional Collisions.

Contest 2 syllabus (9 Oct 2026, 2:17 PM) — VERBATIM:

Forces & Types · Newton's Laws of Motion · Frames of Reference · Pseudo Forces & Inertia · Free Body Diagrams · Friction & Types · Normal & Coefficient of Friction · Multi-D FBDs with Friction · Multi-D NLM & Friction Problems · Angular Coordinate System · Radial and Tangential Vectors · CM: Angular Displacement & Speed · CM: Angular Velocity & Acceleration · CM: Centripetal & Centrifugal · Solving Circular Motion · Kinetic Energy · Potential Energy · Work Done · Spring Force · Conservative vs Non-Conservative Forces · Work and Energy Theorem · Centre of Mass · CoM: System of Objects · CoM: Distributed Systems · CoM: Multi-Dimensional Objects · Linear Momentum · Conservation of Linear Momemtum · Conservation of LM: Multi-Dimensional · Conservation of LM: Multi-Stage · Collisions for Rigid Bodies · Elastic Collision · Inelastic Collision · Coefficient of Restitution · Multi-Dimensional Collisions — "i.e From Worksheet 05 - 10".

Other AP facts from Slack:

- Contest 1: Fri 11 Sep (postponed to 1:30–3:30 PM), offline, classroom Athena. Syllabus = first four lectures (L1–L4).
- Contest 2: Fri 16 Oct, 9–11 AM, classroom Athena, subjective.
- Exam guidelines PDF + seating-plan PDFs shared as files.slack.com links.
- AP Learning Tracker: docs.google.com/spreadsheets/d/1ZOx2WQ11_7dymZ7tux25VkwUmWyXea1pGLauEx-WNIU (AP column).
- Also from Slack: IA-1 syllabi for PSP (programming) and other subjects exist in the same channel (out of scope here).

SOURCE 3 — WHATSAPP HARDCOPY PHOTOS (146 images, ap_media/m_001 … m_146)

Photo inventory (what each photo is)

- **m_004–m_006:** Maths IA-1 paper (BTNE113) — not AP.
- **Lecture notes:** m_007 (Measurement L01), m_015 (Vectors L02), m_032 (Motion L03), m_045–047 (projectile notes), m_001/m_003/m_048/m_052–053/m_055–056 (NLM/forces notes), m_076–078, m_084–085, m_092, m_100, m_105–113 (notes incl. two-stage momentum m_106, 2D momentum m_107), m_117–124, m_133–134, m_136–137.
- **Measurement worksheet:** m_143 (Q10–11), m_144 (Q6–9), m_145/m_146 (Q1–5); older copies m_012, m_014.
- **Calculus/1D worksheet:** m_025 (Q1–4), m_028 (Q11–12), m_029, m_042–044.
- **Projectile:** m_002, m_049. **Relative velocity:** m_050. **NLM:** m_051, m_054, m_057–059.
- **Friction worksheet:** m_140 (Q8–11), m_141 (Q5–7), m_142 (Q1–4).
- **Circular worksheet:** m_081–091 (single questions incl. Q1, Q6, Q11), m_127 (Q27–30), m_128 (Q22–26), m_129 (Q17–21), m_130 (Q7–11), m_131 (Q12–16), m_132 (Q1–6), m_135 (Q27–30).
- **Work & Energy worksheet:** m_116 (Q1–6), m_115 (Q7–12), m_114 (Q13–15).
- **Collisions worksheet:** m_075 (Q1–4), m_079 (Q13–15), m_072–074 (Q5–12 incl. "move together" stick cases).
- **Momentum worksheet:** m_101 (Q14–17), m_102 (Q10–13), m_103 (Q5–9).
- **CoM worksheet:** m_104 (Q1–4).
- **Contest-1 printed paper:** m_061 (Q28b airplane), m_062 (Q28a truck), m_063/m_064 (Q24, Q25), m_065 (Q21 errors, a(t), traffic), m_066 (Q18–20), m_067 (Q13–17), m_068 (Q3–7), m_069 (Q8–12), m_070–071 (covers).
- **Blank:** m_060, m_080.

ORIGINAL QUESTIONS — Worksheet: Measurement (g = 9.8)

1. Importance of measurement in engineering (explain, 3M).
2. Fundamental vs derived quantities; SI units of any 5 fundamental.
3. Write dimensional formulas: velocity, acceleration, force, work, pressure.
4. Scientific notation + order of magnitude of: 0.000456; 7,850,000; 299,792,458; 0.0000000082.
5. Rod measured 25.60 cm, true 25.30 cm: (a) absolute (b) relative (c) percentage error.
6. Disc radius (10.0 ± 0.2) cm: % error in area $A = \pi r^2$.
7. A = (50.0 ± 0.5), B = (20.0 ± 0.2), Z = A/B: max absolute and relative error.
8. Dimensional check of $s = ut + \frac{1}{2}at^2$.
9. (story) SI vs Imperial units caused a mission failure — explain and give lessons.
10. Cube side (5.00 ± 0.02) cm: (a) % error in volume (b) max absolute error.
11. (story) Mars Climate Orbiter: what happened, which units were mixed, lessons.

ORIGINAL QUESTIONS — Worksheet: Calculus & 1D (photos m_025, m_028)

1. A particle moves along a straight line with position function $x(t) = 6t^3 - 3t^2 + 6t + 2$ (x in metres, t in seconds). Find (a) velocity as a function of time (b) acceleration (c) velocity and acceleration at t = 2 s.
2. Evaluate the indefinite integral $\int(6x^4 - 8x^3 + 5x^2 - 4x + 7)\,dx$.

3. Velocity of a particle $v(t) = 6t^2 - 4t + 3$ m/s; if it starts from position $x = 2$ m at $t = 0$, determine its position at $t = 3$ s.

4. Differentiate $y = 3x^4 - 5x^2 + 7 \sin x + 4e^x - 6 \ln x$; hence find dy/dx at $x = 1$.

11. (robotics story) End-effector position $x(t) = 2t^3 - 9t^2 + 12t$. (a) derive $v(t)$ and $a(t)$ (b) times at which the end-effector is stationary (c) explain how velocity/acceleration info is useful in robotic motion planning with a real-world example.

12. (variable force) $F(x) = kx$. (a) derive work done moving $x = 0$ to X by integration (b) explain why integration is needed when force is not constant (c) one engineering application; numerical case $k = 500$ N/m, $X = 0.20$ m: calculate the work.

ORIGINAL QUESTIONS — Worksheet: Friction & NLM (g = 9.8 unless stated)

1. 5 kg, $\mu s = 0.40$: maximum force without moving.

2. 10 kg pulled 35 N, $\mu k = 0.20$: (a) normal force (b) friction (c) net force (d) acceleration.

3. 6 kg, $\mu s = 0.50$, $\mu k = 0.30$, $F = 20$ N: does it move? actual friction?

4. 8 kg, $\mu s = 0.40$, $\mu k = 0.25$, force ramped from 0: (a) force at which it starts moving (b) friction just after (c) acceleration.

5. 5 kg, $F = 40$ N at 30° above horizontal, $\mu k = 0.20$: acceleration.

6. 10 kg, $F = 50$ N at 30° below horizontal (push), $\mu k = 0.25$: acceleration.

7. 10 kg on 30° incline, $\mu k = 0.20$, slides down: acceleration.

8. 5 kg on 37° incline, $\mu s = 0.50$ ($g = 10$, $\sin 37 = 0.6$, $\cos 37 = 0.8$): stays or slides?

9. 4 kg on 30° incline, $F = 30$ N up-parallel, $\mu k = 0.20$: acceleration.

10. 6 kg pulled up 30° incline by $T = 50$ N, $\mu k = 0.15$: acceleration.

11. 5 kg on rough table ($\mu k = 0.20$) + 3 kg hanging, frictionless pulley: acceleration and tension.

ORIGINAL QUESTIONS — Worksheet: Circular Motion

1. $r = 2$ m, arc 6 m: angle. 2. 120° in radians. 3. $\theta = 3\pi/4$, $r = 4$: arc length. 4. $\theta = 60°$, $r = 5$: (x, y) coordinates. 5. $r = 3$, $v = 6$: centripetal acceleration. 6. $r = 2$, $\omega = 5$: (a) v (b) a_c. 7. $a_t = 4$, $r = 2$: α. 8. $r = 4$, $\omega = 3$, $\alpha = 2$: (a) a_t (b) a_r (c) total a. 9. 20 rev in 10 s: (a) θ (b) ω. 10. $T = 0.5$ s: ω. 11. $f = 5$ Hz: (a) ω (b) T. 12. 8π rad in 4 s: average ω and revolutions. 13. From rest, $\alpha = 3$, $t = 5$: (a) ω (b) θ. 14. $\omega_0 = 4$, $\alpha = 3$, $t = 6$: ω, θ. 15. $\omega_0 = 20$, $\alpha = -4$ to rest: (a) t (b) θ. 16. $\theta = 20$, ω: 4→12: α. 17. $m = 2$, $r = 4$, $v = 8$: (a) a_c (b) F_c. 18. 1000 kg, $r = 50$, $v = 10$: centripetal force. 19. 5 kg, $\omega = 4$, $r = 2$: (a) v (b) a_c (c) F_c. 20. (explain) centripetal vs centrifugal force. 21. 4 kg on 2 m string at 6 m/s: tension. 22. 2 kg, 1.5 m, 4 m/s: (a) a_c (b) T. 23. 1200 kg, $r = 40$, $v = 12$: centripetal force. 24. 5 kg, $r = 2$ m, $\mu = 0.4$: max speed without slipping. 25. $r = 5$, $\omega = 2$, $\alpha = 3$: (a) a_t (b) a_r (c) total. 26. 3 kg, $r = 2$, 4→8 m/s in 2 s: (a) a_t (b) a_c (c) total. 27. (comment) $r = 100$, $v = 20$: a_c; what if speed doubles? 28. $r = 0.5$, 120 rpm: (a) ω (b) v (c) a_c. 29. $r = 3$, $\theta(t) = 2t^2 + 3t$: (a) ω at $t = 2$ (b) α (c) v (d) a_t (e) a_c. 30. $r = 2$, $\omega(t) = 3t^2 + 2$: (a) α at $t = 2$ (b) v (c) a_t (d) a_c.

ORIGINAL QUESTIONS — Worksheet: Work & Energy (g = 9.8)

1. 5 kg at 4 m/s: KE. 2. $p = 10$, $m = 2$: KE. 3. 5 kg at 10 m: PE. 4. $F = 80$ N over 8 m along force: work. 5. $F = 100$ N over 5 m at 60°: work. 6. Spring $k = 200$, stretch 0.1 → 0.3 m: work by spring. 7. (explain) same work on two paths → what does it mean for the force? 8. 4 kg at rest, net work 200 J: final speed. 9. 5 kg at 4 m/s, 90 J work: final speed. 10. 2 kg free-falls 20 m from rest: speed at ground. 11. 5 kg falls 8 m, loses 50 J to friction: final speed. 12. 10 kg falls 5 m, loses 100 J to friction: final speed. 13. (5-part) 4 kg pushed 30 N through 10 m against 5 N friction from rest: (a) work by push (b) work by friction (c) net work (d) final KE (e) final speed. 14. KE constant, $m \to 4m$: what happens to v? 15. (explain) fall with friction: is mechanical energy conserved?

ORIGINAL QUESTIONS — Worksheet: Collisions & Momentum

1. 2 kg at 6 m/s head-on into stationary 4 kg: total momentum before. 2. 5 kg at 4 + stationary 3, stick: common velocity. 3. 2 kg at 8 + stationary 6, perfectly inelastic: (a) p_i (b) v (c) KE_i (d) KE_f (e) energy lost. 4. 2 kg at 10 elastic into stationary 3: both final velocities. 5. 4 kg at 6 vs 2 kg at 3 opposite: both final velocities (elastic). 6. 5 kg at 8 + stationary 3, stick: (a) v (b) KE_i (c) KE_f (d) lost. 7. 2 kg at 10, stationary 3, e = 0.6: both v. 8. 4 kg at 8, stationary 6, e = 0.5: both v. 9. 3 kg at 6, stationary 2, e = 0.4: approach speed, separation speed, both v. 10. 2 kg at 12, stationary 4, e = 0.5: both v. 11. 4 kg east at 5 + 6 kg north at 3, stick: final speed and direction. 12. 5 kg at +x 8 + 3 kg at +y 6, stick: final speed and direction. 13. Identical balls elastic, first at 8, second rest: final velocities. 14. Two bodies of equal mass, first at 10 m/s, second at rest, perfectly elastic 1D: final velocities + explain result. 15. 4 kg at 5, 6 kg at 2 same direction, e = 0.5: final velocities. (Printed collisions worksheet ends at Q15 as photographed — it has NO Q16/Q17.)

Momentum worksheet (separate sheet; photos m_103 Q5–9, m_102 Q10–13, m_101 Q14–17 — **the Q1–4 page was never photographed**): 5. Uniform rod 6 m, 12 kg from x = 2 to x = 8: CoM. 6. 5 kg at 4 m/s: momentum. 7. 2 kg right at 8 + 4 kg left at 3: total momentum. 8. 0.5 kg at 10 rebounds 6 from wall: (a) p_i (b) p_f (c) Δp (d) impulse. 9. 4 kg at 6 + stationary 2 stick: v. 10. 2 kg east at 6, 3 kg north at 4: (a) p_x (b) p_y (c) magnitude (d) direction. 11. 2 kg east at 5 + 1 kg north at 4 stick: (a) final components (b) speed (c) direction. 12. 2 kg at 4 + 3 kg at 6 same direction: (a) total momentum (b) centre-of-mass velocity. 13. System mass 20, external force 60 N: acceleration of CoM. 14. Mass 10, a_cm = 2: net force. 15. 2 kg at 10 collides stationary 3 then combined 5 hits stationary 5 (all stick): final velocity. 16. 10 kg at rest explodes into 4 kg at 12 m/s + 6 kg: velocity of 6 kg piece. 17. F_net = 0 on a system: (a) total momentum (b) velocity of centre of mass (c) acceleration of centre of mass.

ORIGINAL QUESTIONS — Worksheet: Centre of Mass

1. 2 kg at x = 0, 8 kg at x = 5: x_cm. 2. 2 kg at (0,0), 3 kg at (4,0), 5 kg at (10,0): x_cm. 3. 2 kg and 8 kg separated by 10 m: CoM measured from the 2 kg mass. 4. 2 kg at (0,0), 3 kg at (6,0), 5 kg at (0,4): CoM point.

Contest-2 (portal): ice cube (m, L) melts in tray (M) — does CoM move, direction, effect of mass ratio (no numbers).

ORIGINAL QUESTIONS — CONTEST-1 printed paper (11 Sep, offline)

Part A (1M MCQ): Q1–2 (page cover, definitions). Q3 Which error from unpredictable variations (options: gross/systematic/random/percentage). Q4 Body from rest, a = 2 m/s², velocity after 5 s (5/10/15/20). Q5 Which is a vector quantity (mass/time/speed/velocity). Q6 Direction of cross product determined by (triangle law/parallelogram law/right-hand rule/head-to-tail). Q7 Rate of change of velocity with time (displacement/speed/acceleration/jerk). Q8 Maximum range of projectile (30/45/60/90). Q9 Significant figures in 0.00456. Q10 SI unit of acceleration. Q11 Vectors of magnitude 5 and 4 perpendicular: dot product. Q12 u = 10, v = 30 in 5 s: average acceleration. Q13 Vector multiplied by negative scalar: what changes.

Part B (2M, answer all): Q16 Order of magnitude: definition + 0.00000487 and 564000000. Q17 Dot vs cross product, at least two differences each. Q18 (1) dy/dx of $y = 4x^2 - 3x + 7$ (2) $\int(3x^2 + 2x)\,dx$. Q19 Two position-time graphs (straight line; upward curve): state how average and instantaneous velocity change in each. Q20 Car A 20 m/s, Car B 12 m/s same direction: velocity of A rel B; then if opposite directions.

Part C (5M): Q21 F = 80.0 ± 0.8 N, a = 25.0 ± 0.5 m/s²: (1) mass (2) max error (3) relative error (4) percentage error. Q22–Q23 (pages not photographed in this set — unknown). Q24 Ball kicked 25 m/s at 37° (g = 10, sin37 = 0.6, cos37 = 0.8): (1) components (2) time to return (3) max height. Q25 Drone: $x(t) = 3t^2 + 4t + 2$, $y(t) = 2t^3 - 3t^2 + 5t$: (1) velocity components and speed at t = 2 (2) acceleration components and magnitude at t = 2 (3) direction of motion vs +x axis. Q26–Q27 (not in photographed set). Unnumbered 5M: a(t) = 10t + 6 with v(0) = 4, x(0) = 3: (1) v(t) (2) x(t) (3) x(1). Unnumbered 5M traffic story: 0→20 m/s in 5 s then constant 20 for 5 s: v-t graph, displacement, average velocity, average acceleration first 5 s.

Part D (10M, answer ONE): Q28a Ball-in-truck: person in back of truck throws ball vertically up at 5.0 m/s relative to truck; truck moves forward at 10.0 m/s with constant acceleration 2.0 m/s²; g = 10; hand fixed relative to truck;

catch at same height. (a) Evaluate claim: "ball returns because it already has the truck's horizontal velocity" — correct? (b) Support: initial velocity vector rel ground, time to return, horizontal separation from hand, ahead or behind? (c) Design: horizontal component of launch rel truck (keep upward 5.0) needed to return to hand; does design change if truck's initial speed doubles with same acceleration? (d) Check: significant digits in the time from (b). Q28b Airplane in wind: 100 km/h through air pointing due north, wind 60 km/h east, east = +î north = +ĵ. (a) Evaluate claim: "point due north → travels due north over ground" — correct? (b) Support: vectors for air velocity and wind, sum = ground velocity, magnitude and direction, displacement after 30 minutes, sketch triangle. (c) Design: required east–west and north–south components of air velocity to travel due north (speed still 100), ground speed, sketch direction (no angle needed). (d) Assess limitation: wind 120 km/h with airspeed 100 — can any heading make it go due north? Justify with components.

WHAT IS NOT IN THIS FILE

- **Never photographed (physically absent from the 146-photo set — verified 26 Oct, all 146 classified):** Contest-1 Q1–2, Q14–15 (numbering gap before Part-B Q16), Q22–23, Q26–27 (covers m_070/m_071 are S-VYASA USN answer sheets, not question pages); Momentum worksheet Q1–4 (first page); rotational worksheet anything after Q13 (pages not in set); printed bank-A pages 1–2 (worksheet starts at Q1 on p3 — likely just title pages).
- **Partially cut:** m_042 Q10 top edge cut off in photo.
- Old dim photos m_012/m_014 are earlier shots of the same measurement worksheet now transcribed verbatim from m_143–146.
- Duplicate photos: m_054 (= NLM Q1–5), m_058/m_059 (same NLM page), m_072 (partial = m_079 collisions Q13–15), m_091 (= rotational Q1–5), m_127/m_135 (same bank-A page 8), m_133/m_134 & m_076–078 etc. = lecture-notes pages, no questions.
- The 146 photos themselves live in ap_media/m_001 … m_146 (not public; ask for specific pages).
- Slack live session expired (QR re-login needed); all AP announcements already captured above.

NEWLY TRANSCRIBED — EVERY REMAINING PHOTO

Worksheet: Projectile (photos m_002, m_049)

6. A projectile is launched with an initial speed of 20 m/s at an angle of 30° above the horizontal. Determine its time of flight and horizontal range. Take g = 9.8 m/s².

7. A projectile is fired with a speed of 40 m/s at an angle of 45°. Determine its maximum height and horizontal range. Take g = 9.8 m/s².

8. A particle is projected with an initial velocity of 30 m/s at an angle of 60° with the horizontal. Determine the horizontal and vertical components of its initial velocity.

9. A velocity vector has components vx = 12 m/s, vy = 5 m/s. Determine its magnitude and direction with respect to the positive x-axis.

10. A particle has initial velocity components vx0 = 10 m/s, vy0 = 20 m/s. The acceleration components are ax = 2 m/s², ay = –9.8 m/s². Determine its position components after 3 s, assuming it starts from the origin.

11. A ball is projected from the ground with a velocity of 25 m/s at an angle of 37°. Determine: (a) initial horizontal velocity (b) initial vertical velocity (c) time of flight (d) maximum height (e) horizontal range. Take $\sin 37° = 0.6$, $\cos 37° = 0.8$, $g = 10\ m/s^2$.

12. Two particles A and B are initially separated by 500 m. Particle A moves with a velocity of 20 m/s while particle B moves toward A with a velocity of 30 m/s. Determine: (a) relative velocity of A with respect to B (b) time taken for them to meet.

Worksheet: Relative Velocity + 1D basics (photo m_050)

1. Two cars A and B are moving in the same direction with velocities 25 m/s and 15 m/s, respectively. Determine the velocity of car A relative to car B.

2. Two trains are moving in opposite directions with speeds 20 m/s and 30 m/s. Find their relative velocity.

3. A particle starts from rest and moves with a constant acceleration of 4 m/s^2 for 8 s. Determine its final velocity and displacement.

4. A car travelling at 20 m/s accelerates uniformly at 3 m/s^2. Find its velocity after 6 s and the distance travelled during this time.

5. A stone is dropped from a height and reaches the ground after 4 s. Taking $g = 9.8\ m/s^2$, determine the height from which it was dropped.

Worksheet: Calculus in motion (photos m_025, m_028, m_029)

1. A particle moves along a straight line according to the position function $x(t) = 6t^3 - 3t^2 + 6t + 2$, where x is in metres and t in seconds. Find: (a) the velocity of the particle as a function of time (b) the acceleration of the particle (c) the velocity and acceleration at $t = 2$ s.

2. Evaluate the indefinite integral: $\int(6x^4 - 8x^3 + 5x^2 - 4x + 7)\,dx$.

3. The velocity of a particle is given by $v(t) = 6t^2 - 4t + 3$ m/s. If the particle starts from the position $x = 2$ m at $t = 0$, determine its position at $t = 3$ s.

4. Differentiate the following function with respect to x: $y = 3x^4 - 5x^2 + 7 \sin x + 4e^x - 6 \ln x$. Hence find the value of dy/dx at $x = 1$.

8. The velocity of a vehicle is described by v(t) = 10e^(–0.2t) + 2 m/s. Determine the total displacement of the vehicle during the first 10 s.

9. A particle has acceleration $a(t) = 12t^2 - 6t + 4\ m/s^2$. Its initial velocity is $v(0) = 5$ m/s and its initial position is $x(0) = 2$ m. Using integration, determine: (a) the velocity function v(t) (b) the position function x(t) (c) the position of the particle at $t = 2$ s.

10. A rectangular sheet has a fixed perimeter of 40 cm. If its length is x cm, express its area A as a function of x. Using differentiation: (a) determine the value of x for which the area is maximum (b) determine the corresponding width (c) calculate the maximum possible area.

11. In robotics, the position of a robot's end-effector can be represented as a time-dependent function. Consider a simplified one-dimensional model: $x(t) = 2t^3 - 9t^2 + 12t$. Investigate how differentiation can be used to determine the velocity and acceleration of the end-effector. Using the given function: (a) derive v(t) and a(t) (b) determine the times at which the end-effector is stationary (c) explain how velocity and acceleration information can be useful in robotic motion planning and control. Support your explanation with at least one real-world robotics example.

12. Variable forces are encountered in engineering systems such as springs, robotic actuators, and mechanical mechanisms. Consider a force that varies with displacement according to $F(x) = kx$, where k is a constant. Using integration: (a) derive an expression for the work done in moving an object from $x = 0$ to $x = X$ (b) explain why integration is required when the force is not constant (c) investigate one engineering application where the work done by a variable force is calculated using integration. For the numerical case $k = 500$ N/m, $X = 0.20$ m, calculate the work done.

Worksheet: 1D Motion basics (photo m_043)

1. A car travels 150 m in 10 s along a straight road. Determine its average speed in m/s and km/h.

2. A student walks 120 m east and then 50 m west along the same straight path. Determine the total distance travelled and the displacement of the student.

3. A car starts from rest and moves with a constant acceleration of 3 m/s^2 for 8 s. Determine its final velocity and the displacement covered during this interval.

4. A particle has an initial velocity of 12 m/s and accelerates uniformly at 2.5 m/s^2 for 6 s. Determine its final velocity and displacement.

5. The velocity of a particle changes uniformly from 8 m/s to 32 m/s in 6 s. Determine the acceleration and displacement of the particle during this interval.

Worksheet: calculus/1D continued (photo m_042; top edge cut on Q10)

10. The acceleration of a particle varies uniformly from 2 m/s^2 at t = 0 to (value cut off) at t = … . Determine the change in velocity and final velocity using the area under the acceleration-time graph.

11. A robot's end-effector moves along a straight axis according to $x(t) = 2t^3 - 9t^2 + 12t$, where x is in metres and t is in seconds. Determine the velocity, acceleration, and jerk as functions of time. Also determine the instants at which the end-effector is momentarily at rest and the displacement between those instants.

12. A particle moves along a straight line with acceleration $a(t) = 4t - 2$, where a is in m/s^2 and t is in seconds. At t = 0, its velocity is 3 m/s and its position is 2 m. Determine the velocity function, position function, velocity at t = 4 s, displacement during the first 4 s, and jerk.

Worksheet: 1D/calculus continued (photo m_044)

6. The position of a particle moving along a straight line is given by $x(t) = 3t^3 - 6t^2 + 4t + 2$, where x is in metres and t is in seconds. Determine the velocity and acceleration at t = 2 s.

7. A particle moves with an initial velocity of 20 m/s and a constant acceleration of -4 m/s^2. Determine the time at which it comes momentarily to rest and the displacement covered before coming to rest.

8. A velocity-time graph consists of a straight line increasing from 0 m/s at t = 0 to 24 m/s at t = 6 s, followed by a constant velocity of 24 m/s for the next 4 s. Determine the total displacement and the average velocity over the entire 10 s interval.

9. The velocity of a particle is given by $v(t) = 4t^2 - 6t + 5$, where v is in m/s and t is in seconds. Determine the acceleration at t = 3 s and the displacement of the particle from t = 0 to t = 3 s.

Worksheet: NLM & Friction (photo m_051)

1. A 5 kg block rests on a horizontal frictionless surface. A horizontal force of 20 N is applied to the block. Determine its acceleration.

2. A 10 kg object is subjected to two horizontal forces of 35 N and 15 N acting in opposite directions. Determine the net force and acceleration of the object.

3. A 4 kg block is pulled along a horizontal surface by a force of 25 N. The coefficient of kinetic friction between the block and surface is 0.20. Determine: (a) normal force (b) frictional force (c) net force (d) acceleration. Take g = 9.8 m/s^2.

4. A person of mass 60 kg stands on a weighing machine inside a stationary elevator. Determine the reading of the weighing machine. Take g = 9.8 m/s^2.

5. A 50 kg person stands inside an elevator accelerating upward at 2 m/s^2. Determine the normal force exerted by the elevator floor on the person. Take g = 9.8 m/s^2.

Worksheet: NLM continued (photo m_057; m_054 is a duplicate of the Q1–5 page)

6. A 50 kg person stands inside an elevator accelerating downward at 3 m/s^2. Determine the apparent weight of the person. Take g = 9.8 m/s^2.

7. A car of mass 1000 kg experiences a driving force of 4000 N and a resistive force of 1000 N. Determine: (a) net force (b) acceleration (c) velocity after 8 s if it starts from rest.

8. A 10 kg block is placed on a horizontal surface with coefficient of static friction 0.40. A horizontal force is gradually increased from zero. Determine the maximum horizontal force that can be applied without causing the block to move. Take g = 9.8 m/s^2.

9. A 5 kg block is pulled by a force of 30 N at an angle of 30° above the horizontal on a rough horizontal surface. The coefficient of kinetic friction is 0.20. Determine: (a) horizontal component of the applied force (b) vertical component (c) normal reaction (d) frictional force (e) acceleration. Take g = 9.8 m/s^2.

Worksheet: NLM incline & connected blocks (photos m_058 + m_059 — same page photographed twice)

10. A 10 kg block is placed on a frictionless inclined plane making an angle of 30° with the horizontal. Determine: (a) component of weight parallel to the plane (b) component of weight perpendicular to the plane (c) normal reaction (d) acceleration of the block. Take g = 9.8 m/s^2.

11. A 6 kg block is placed on a rough inclined plane making an angle of 30° with the horizontal. The coefficient of kinetic friction is 0.20. Determine the acceleration of the block as it slides down the plane. Take g = 9.8 m/s^2.

12. Two blocks of masses 4 kg and 6 kg are connected by a light inextensible string on a frictionless horizontal surface. A horizontal force of 30 N is applied to the 6 kg block. Determine: (a) acceleration of the system (b) tension in the string.

13. A 20 kg block is pulled on a horizontal surface by a force of 100 N acting at 37° above the horizontal. The coefficient of kinetic friction is 0.25. Take sin37° = 0.6, cos37° = 0.8, g = 10 m/s^2. Determine the acceleration of the block.

VERBATIM FULL TEXT — PRINTED WORKSHEETS (every question exactly as printed, 26 Oct page-by-page reading)

FRICTION WORKSHEET — full text (photos m_142 Q1–4, m_141 Q5–7, m_140 Q8–11; header "Worksheet Questions")

1. A 5 kg block is placed on a horizontal rough surface. The coefficient of static friction is 0.40. Determine the maximum horizontal force that can be applied without causing the block to move. Take g = 9.8 $m\ s^{-2}$.

2. A 10 kg block is pulled horizontally by a force of 35 N. The coefficient of kinetic friction between the block and the surface is 0.20. Determine: (a) Normal force (b) Kinetic friction (c) Net horizontal force (d) Acceleration

3. A 6 kg block is initially at rest on a horizontal surface. The coefficients of static and kinetic friction are 0.50 and 0.30 respectively. A horizontal force of 20 N is applied. Determine whether the block moves. If it does not move, determine the actual static friction acting on the block. Take g = 9.8 m s^{-2}.

4. A 8 kg block is on a horizontal rough surface. The coefficient of static friction is 0.40 and the coefficient of kinetic friction is 0.25. A horizontal force is gradually increased from zero. Determine: (a) The force at which the block just begins to move. (b) The frictional force immediately after motion begins, assuming the applied force remains equal to the limiting value. (c) The acceleration at that instant. Take g = 9.8 m s^{-2}.

5. A 5 kg block is pulled by a force of 40 N at an angle of 30° above the horizontal on a rough horizontal surface. The coefficient of kinetic friction is 0.20. Determine: (a) Horizontal component of the applied force (b) Vertical component of the applied force (c) Normal force (d) Kinetic friction (e) Acceleration of the block Take g = 9.8 m s^{-2}.

6. A 10 kg block is pushed by a force of 50 N at an angle of 30° below the horizontal on a rough horizontal surface. The coefficient of kinetic friction is 0.25. Determine: (a) Normal force (b) Frictional force (c) Net horizontal force (d) Acceleration Take g = 9.8 m s^{-2}.

7. A 10 kg block is placed on a rough inclined plane making an angle of 30° with the horizontal. The coefficient of kinetic friction is 0.20. Determine: (a) Component of weight parallel to the plane (b) Component of weight perpendicular to the plane (c) Normal reaction (d) Frictional force (e) Acceleration of the block as it slides down Take g = 9.8 m s^{-2}.

8. A 5 kg block is placed on a rough inclined plane of angle 37°. The coefficient of static friction is 0.50. Determine whether the block remains at rest or starts sliding. Take sin 37° = 0.6, cos 37° = 0.8, g = 10 m s^{-2}.

9. A 4 kg block is on a rough inclined plane of angle 30°. A force of 30 N is applied parallel to the plane upward. The coefficient of kinetic friction is 0.20. Determine the acceleration of the block and state its direction. Take g = 9.8 m s^{-2}.

10. A 6 kg block is pulled up a rough inclined plane of angle 30° by a tension of 50 N parallel to the plane. The coefficient of kinetic friction is 0.15. Draw the complete FBD and determine the acceleration of the block. Take g = 9.8 m s^{-2}.

11. A 5 kg block rests on a horizontal rough surface. It is connected by a light string to a hanging mass of 3 kg over a frictionless pulley. The coefficient of kinetic friction between the 5 kg block and the horizontal surface is 0.20. Draw the FBD of both masses and determine: (a) Acceleration of the system (b) Tension in the string Take g = 9.8 m s^{-2}.

CIRCULAR WORKSHEET (printed bank A, "Worksheet Questions", pages 3–8) — full text (photos m_132 Q1–6, m_131 Q7–11, m_130 Q12–16, m_129 Q17–21, m_128 Q22–26, m_127 = m_135 Q27–30)

1. A particle moves along a circular path of radius 2 m and covers an arc length of 6 m. Determine its angular displacement in radians.

2. A wheel rotates through an angle of 120°. Express this angular displacement in radians.

3. A particle moves through an angular displacement of 3π/4 radians on a circular path of radius 4 m. Find the distance traveled along the circular path.

4. A particle is located at an angular position of 60° on a circular path of radius 5 m. Determine its Cartesian coordinates (x, y).

5. A particle moves in a circle of radius 3 m with a speed of 6 m/s. Determine its radial acceleration.

6. A particle moves in a circular path of radius 2 m with angular velocity 5 rad/s. Find: (a) linear velocity, (b) radial acceleration.

7. A particle has a tangential acceleration of 4 m/s^2 while moving in a circular path of radius 2 m. Determine its angular acceleration.

8. A particle moves in a circular path with r = 4 m, ω = 3 rad/s and α = 2 rad/s^2. Determine: (a) tangential acceleration, (b) radial acceleration, (c) magnitude of total acceleration.

9. A wheel rotates through 20 revolutions in 10 seconds. Determine: (a) total angular displacement, (b) average angular speed.

10. A wheel has a time period of 0.5 s. Determine its angular speed.

11. A rotating disc has a frequency of 5 Hz. Determine: (a) angular speed, (b) time period.

12. A particle moves through an angular displacement of 8π radians in 4 seconds. Determine its average angular speed and number of revolutions completed.

13. A wheel starts from rest and rotates with a constant angular acceleration of 2 rad/s² for 5 seconds. Determine: (a) final angular velocity, (b) angular displacement.

14. A rotating disc has an initial angular velocity of 4 rad/s and angular acceleration of 3 rad/s². Determine its angular velocity after 6 seconds.

15. A wheel rotating at 20 rad/s is brought to rest with a constant angular acceleration of -4 rad/s². Determine: (a) time required to stop, (b) angular displacement before stopping.

16. A particle rotates through an angular displacement of 20 rad while its angular velocity increases from 4 rad/s to 12 rad/s. Determine its angular acceleration.

17. A 2 kg object moves in a circular path of radius 4 m with a speed of 8 m/s. Determine: (a) centripetal acceleration, (b) centripetal force.

18. A car of mass 1000 kg moves around a circular track of radius 50 m at 10 m/s. Determine the required centripetal force.

19. A 5 kg object moves in a circle with angular velocity 4 rad/s and radius 2 m. Determine: (a) linear velocity, (b) centripetal acceleration, (c) centripetal force.

20. Explain the difference between centripetal force and centrifugal force. Give one real-life example of each concept.

21. A 4 kg object is attached to a string of length 2 m and moves in a horizontal circle at a speed of 6 m/s. Determine the tension in the string assuming that the tension provides the required centripetal force.

22. A 2 kg stone is tied to a string of length 1.5 m and rotated in a horizontal circle at 4 m/s. Determine: (a) centripetal acceleration, (b) tension in the string.

23. A car of mass 1200 kg travels around a circular curve of radius 40 m at 12 m/s. Determine the frictional force required to provide the centripetal force.

24. A 5 kg block is moving in a horizontal circle of radius 2 m. If the coefficient of friction between the block and the surface is 0.4, determine the maximum speed with which the block can move without slipping. Take $g = 9.8$ m/s².

25. A particle moves in a circular path of radius 5 m. Its angular velocity is 2 rad/s and angular acceleration is 3 rad/s². Determine: (a) tangential acceleration, (b) radial acceleration, (c) total acceleration.

26. A 3 kg object moves in a circular path of radius 2 m. Its speed increases from 4 m/s to 8 m/s in 2 seconds. Determine: (a) average tangential acceleration, (b) centripetal acceleration at 8 m/s, (c) total acceleration at the final speed.

27. A car moves around a circular track of radius 100 m. Its speed is 20 m/s, determine the centripetal acceleration. If the speed is doubled, determine the new centripetal acceleration. Comment on the effect of speed on centripetal acceleration.

28. A wheel of radius 0.5 m rotates at 120 rpm. Determine: (a) angular speed in rad/s, (b) linear speed at the rim, (c) centripetal acceleration at the rim.

29. A particle moves in a circle of radius 3 m with angular position $\theta(t) = 2t^2 + 3t$ where θ is in radians and t is in seconds. Determine: (a) angular velocity at $t = 2$ s, (b) angular acceleration, (c) linear velocity at $t = 2$ s, (d) tangential acceleration, (e) centripetal acceleration at $t = 2$ s.

30. A particle moves in a circular path of radius 2 m with angular velocity $\omega(t) = 3t^2 + 2$. Determine: (a) angular acceleration at $t = 2$ s, (b) linear velocity at $t = 2$ s, (c) tangential acceleration at $t = 2$ s, (d) centripetal acceleration at $t = 2$ s.

ROTATIONAL WORKSHEET (SECOND printed sheet, own numbering, "Worksheet Questions") — full text (photos m_089 =

m_091 Q1–5, m_088 (p4) Q6–10, m_087 (p5) Q11–13; no page after Q13 exists in the photo set)

1. A car moves around a circular track of radius 50 m at a constant speed of 10 m/s. Determine its angular velocity.

2. A wheel of radius 0.4 m rotates with angular velocity 8 rad/s. Determine the linear velocity of a point on its rim.

3. A wheel rotates about its central axis. A point on the rim is 0.3 m from the axis. If the angular velocity is 10 rad/s, determine its linear speed.

4. A rotating disc has radius 2 m and angular velocity 4 rad/s. Determine the linear velocity at: (a) the centre, (b) a point 1 m from the centre, (c) a point on the rim.

5. A wheel rotates through 180° in 2 seconds. Determine its average angular velocity.

6. A wheel of radius 0.5 m rotates at 120 rpm. Determine: (a) angular velocity in rad/s, (b) linear velocity of a point on the rim.

7. The angular position of a rotating body is given by $\theta(t) = 3t^2 + 2t$ where θ is in radians and t is in seconds. Determine: (a) angular position at t = 2 s, (b) angular velocity at t = 2 s.

8. A wheel starts from rest and reaches an angular velocity of 20 rad/s in 5 seconds. Determine its angular acceleration.

9. A rotating disc has initial angular velocity 5 rad/s and angular acceleration 3 rad/s². Determine its angular velocity after 4 seconds.

10. A force of 20 N is applied perpendicular to a spanner at a distance of 0.25 m from the bolt. Determine the torque.

11. A force of 100 N is applied at a distance of 0.5 m from an axis. Determine the torque when the force makes: (a) 90° with the radius, (b) 60° with the radius, (c) 0° with the radius.

12. A wheel starts from rest with a constant angular acceleration of 2 rad/s². Determine: (a) angular velocity after 6 seconds, (b) angular displacement after 6 seconds.

13. A disc of radius 0.5 m starts from rest and rotates with constant angular acceleration of 4 rad/s² for 5 seconds. Determine: (a) final angular velocity, (b) angular displacement, (c) linear velocity at the rim, (d) tangential acceleration at the rim.

WORK & ENERGY WORKSHEET — full text (photos m_116 (p7) Q1–6, m_115 (p8) Q7–12, m_114 Q13–15)

1. A body of mass 5 kg moves with a velocity of 4 m/s. Calculate its kinetic energy.

2. A particle of mass 2 kg has momentum 10 kg m/s. Determine its kinetic energy using $K = p^2/2m$.

3. A 5 kg object is raised to a height of 10 m. Calculate its gravitational potential energy. Take g = 9.8 m/s².

4. A constant horizontal force of 50 N moves a box through a distance of 8 m in the direction of the force. Determine the work done.

5. A force of 100 N acts on an object through a displacement of 5 m at an angle of 60° to the direction of displacement. Calculate the work done.

6. A spring is stretched from 0.1 m to 0.3 m. If its spring constant is 200 N/m, determine the work done by the spring.

7. A block moves from point A to point B along two different paths. The work done by a particular force is the same along both paths. Is the force conservative? Explain.

8. A 4 kg object initially at rest is acted upon by a net force that does 200 J of work. Determine its final speed.

9. A 5 kg block has an initial speed of 4 m/s. If the net work done on it is 90 J, determine its final speed.

10. A 2 kg object is released from rest at a height of 20 m. Neglecting air resistance, determine its speed just before reaching the ground. Take g = 9.8 m/s².

11. A 5 kg block slides down from a height of 8 m to the ground. During the motion, friction performs –50 J of work. If the block starts from rest, determine its final kinetic energy. Take g = 9.8 m/s².

12. A 10 kg block is initially at rest at a height of 5 m. Friction removes 100 J of mechanical energy. Determine the speed of the block when it reaches the ground. Take g = 9.8 m/s².

13. A 4 kg block is pushed along a horizontal surface by a constant force of 30 N through a distance of 10 m. Friction opposing the motion is 5 N. If the block starts from rest, determine: (a) work done by the applied force, (b) work done by friction, (c) net work, (d) final kinetic energy, (e) final speed.

14. A block of mass m is moving with velocity v and has kinetic energy K. If its mass is increased to 4m and its kinetic energy remains unchanged, determine its new velocity in terms of v.

15. A particle moves from position A to position B under the action of gravity and friction. Explain how the work done by gravity and friction affects the mechanical energy of the particle.

CENTRE OF MASS WORKSHEET — full text (photo m_104 (p10); margin note on page: "Same, just (x,y) → the CM values ($\bar{x}$, $\bar{y}$)")

1. Two particles of masses 2 kg and 8 kg are located at x = 0 m and x = 5 m respectively. Determine the position of the centre of mass.

2. Three particles have masses and positions m_1 = 2 kg, x_1 = 0 m; m_2 = 3 kg, x_2 = 4 m; m_3 = 5 kg, x_3 = 10 m. Determine the centre of mass of the system.

3. Two objects of masses 2 kg and 8 kg are separated by 10 m. Determine the position of the centre of mass measured from the 2 kg mass.

4. Three particles of masses 2 kg, 3 kg and 5 kg are located at (0,0), (6,0), (0,4) respectively. Determine the centre of mass of the system.

COLLISIONS WORKSHEET — full text (photos m_075 Q1–4, m_074 (p4) Q5–8, m_073 Q9–12, m_079 Q13–15; m_072 = partial duplicate of the Q13–15 page; sheet ends at Q15)

1. A 2 kg ball moving with a velocity of 6 m s^{-1} collides head-on with a stationary 4 kg ball. Determine the total momentum of the system before collision.

2. A 5 kg block moving at 4 m s^{-1} collides with a stationary 3 kg block. The two blocks stick together after collision. Determine their common velocity after collision.

3. A 2 kg body moving at 8 m s^{-1} collides with a stationary 6 kg body. The collision is perfectly inelastic. Determine: (a) Initial momentum (b) Final common velocity (c) Initial kinetic energy (d) Final kinetic energy (e) Kinetic energy lost

4. A 2 kg ball moving at 10 m s^{-1} collides elastically with a stationary 3 kg ball. Determine the final velocities of both balls.

5. A 4 kg block moving at 6 m s^{-1} collides elastically with a 2 kg block moving in the opposite direction at 3 m s^{-1}. Take the positive direction as the initial direction of motion of the 4 kg block. Determine the final velocities of both blocks.

6. A 5 kg object moving at 8 m s^{-1} collides with a stationary 3 kg object. After collision, the two objects move together. Determine: (a) Common velocity (b) Initial kinetic energy (c) Final kinetic energy (d) Kinetic energy lost

7. A 2 kg ball moving at 10 m s^{-1} collides with a stationary 3 kg ball. The coefficient of restitution is 0.6. Determine the velocities of the two balls after collision.

8. A 4 kg block moving at 8 m s^{-1} collides with a stationary 6 kg block. The coefficient of restitution is 0.5. Determine the final velocities of both blocks.

9. A 3 kg ball moving at 6 m s^{-1} collides with a stationary 2 kg ball. The coefficient of restitution is 0.4. Determine: (a) Velocity of each ball after collision (b) Relative speed of separation (c) Relative speed of approach

10. A 2 kg object moving at 12 m s^{-1} collides with a stationary 4 kg object. The coefficient of restitution is 0.5. Determine the final velocities using: (a) Conservation of momentum (b) Coefficient of restitution

11. A 4 kg object moves east at 5 m s^{-1} and collides with a 6 kg object moving north at 3 m s^{-1}. The objects stick together. Determine the magnitude and direction of the velocity of the combined object after collision.

12. A 5 kg block moving at 8 m s^{-1} along the positive x-axis collides with a 3 kg block moving at 6 m s^{-1} along the positive y-axis. The blocks stick together. Determine the magnitude and direction of their common velocity.

13. Two identical balls collide elastically in one dimension. The first ball has an initial velocity of 8 m s^{-1} and the second ball is initially at rest. Determine the final velocities of both balls.

14. Two bodies of equal mass undergo a perfectly elastic one-dimensional collision. The first body moves at 10 m s^{-1} while the second body is at rest. Determine the final velocities and explain the result.

15. A 4 kg block moving at 5 m s^{-1} collides with a 6 kg block moving at 2 m s^{-1} in the same direction. The coefficient of restitution is 0.5. Determine the final velocities of the two blocks.

PHOTO AUDIT (all 146 classified, 26 Oct 2026)

- Question pages: m_002, m_012, m_014, m_025, m_028, m_029, m_042, m_043, m_044, m_049, m_050, m_051, m_054(dup), m_057, m_058, m_059(dup), m_062–m_069 (CONTEST-1 paper pages: Q28a, Q24, Q21, Q18–19, Q13, Q3, Q8–12), m_072(dup), m_073, m_074, m_075, m_079, m_087, m_088, m_089, m_091(dup), m_101, m_102, m_103, m_104, m_114, m_115, m_116, m_127, m_128, m_129, m_130, m_131, m_132, m_135(dup), m_140, m_141, m_142, m_143–m_146.
- Exam answer sheets / OMR covers (no questions): m_004, m_005, m_006, m_070, m_071.
- Lecture-notes pages (no questions): m_001, m_003, m_007, m_015, m_032, m_045–m_047, m_048, m_052, m_053, m_055, m_056, m_076, m_077, m_078, m_084, m_085, m_090, m_092, m_100, m_105, m_106, m_107, m_108, m_109, m_110, m_111, m_112, m_113, m_117–m_126, m_133, m_134, m_136, m_137.
- Printed textbook/study-material pages & handwritten scratch (no worksheet questions): m_008–m_011, m_013, m_016–m_024, m_026, m_027, m_030, m_031, m_033–m_041, m_093–m_099, m_139.
- Blank / bad shots: m_060, m_061, m_080, m_086, m_095, m_096, m_138.